

No. of Functional Features Included in the Solution

Date	1 st July 2026
Team ID	XXXXXX
Project Name	Credit card approval prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Multi-Table Data Integration	Merges applicant demographic details and rolling transactional histories using clean index matching on customer IDs.	Data Preprocessing Pipeline	Done	Core / High
2	Missing Categorical Imputation	Replaces missing string values in sparse columns (such as the 30.5% gap in Occupation Type) with custom placeholders to preserve records.	Data Preprocessing Pipeline	Done	Core / High
3	Dynamic Target Binarization	Compresses chronological risk records into explicit binary status targets based on past-due monthly limits.	Data Preprocessing Pipeline	Done	Core / High
4	Minority-Class Weighting	Configures explicit balancing factors into the random forest algorithm to prevent class distribution bias during model building.	Machine Learning Engine	Done	Core / High
5	Pipeline Binary Serialization	Wraps and saves feature parameters and classification rules into a static binary asset for real-time model loading.	Machine Learning Engine	Done	Core / Medium
6	Dynamic HTTP Request Router	Captures incoming form payload dictionaries, parses values, and maps strings synchronously through inference logic paths.	Flask Backend Server	Done	Core / High

7	Responsive Profile Input Form	Provides structured dropdown selection items and numerical input text blocks for entering profile information.	Bootstrap Frontend UI	Done	Core / Medium
8	Dynamic Result Template Renderer	Renders definitive risk verdict text views alongside navigation buttons based on prediction return arguments.	Bootstrap Frontend UI	Done	Core / Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Responsive Input Form, Dynamic Result Screen
2	Backend / Logic	3	Dynamic HTTP Request Router, Class Weighting Logic, Pipeline Serialization
3	Database / Storage	3	Multi-Table Index Matching, Structural String Imputer, Target Binarization Mapping
4	API / Integration	0	None (<i>Project relies entirely on localized inference scripts</i>)
5	Security / Authentication	0	None (<i>Localized development sandbox instance environment</i>)